Seen but Barely Used: Concept-Level Interpretability of Pedestrian Perception in a Sparse End-to-End Driving Model

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***Abstract*—Planning and closed-loop scores do not show whether an end-to-end driving model perceives pedestrians, what it encodes about them, or whether its plan depends on them. We open the object queries of SparseDrive, a sparse query-based end-to-end model, and trace pedestrians from image evidence to the planner on nuScenes, NAVSIM and Bench2Drive/CARLA. The pedestrian class is a nearly orthogonal direction in query space; a sparse autoencoder spreads it over about 20 units whose subspace, but not whose individual units, recurs across training seeds. Walking towards the ego path or standing on a zebra crossing is linearly encoded beyond geometric confounds, yet such pedestrians receive no more planner attention than others. Occlusion lowers the pedestrian logit at matched distance and size, occluded pedestrians are recovered through the query memory, and newly appearing pedestrians are rarely detected (first-sighting recall 0.11 occluded, 0.22 visible). Planner attention ranks the agents whose removal moves the plan (rank-AUC 0.90), but removing a pedestrian from attention and collision rescoring moves the plan by centimetres, and attention masking moves it less than for a matched car. Removing all detected pedestrians from attention, rescoring and planner memory in every frame changes the plan about a sixth as much as removing all vehicles; on NAVSIM this removal leaves the driving score unchanged for pedestrians (EPDMS +0.001) but lowers it for vehicles (-0.023). The pipeline uses only public code and data, and we discuss the implications for safety argumentation.**

***Index Terms*—Autonomous driving, explainable AI, interpretability, pedestrian detection, sparse autoencoders, concept activation vectors.**

I. Introduction

END-TO-END (E2E) driving models map camera images to a planned trajectory and are compared by open-loop displacement error, by the NAVSIM score [1] or by closed-loop driving scores [2]. These numbers say how well a model drives on average. They do not say whether it perceived a pedestrian, what it encoded about that pedestrian, or whether the plan would change if the pedestrian were not there. For vulnerable road users this is the question a safety case under frameworks such as SOTIF [3] or UL 4600 [4] must answer, and it requires looking inside the model [5], [6].

Recent work approaches E2E models from two sides. Planner-level concept discovery learns a sparse dictionary on the latent space of a trajectory-scoring planner and edits it to change driving behaviour [7]. Input-level audits remove objects from images or test for spurious cues and measure the plan change [8]–[10]; on NAVSIM, planners rarely react to a removed pedestrian [8]. Neither side asks how the *perception* part of the model represents a pedestrian, which of its properties it encodes, and through which routes that representation reaches the plan.

Query-based E2E models make this tractable. In SparseDrive [11] every road user is one object query, a 256-dimensional feature refined over six decoder layers, carried to the next frame and read by the planner. We use these queries as the unit of analysis and ask five research questions:

RQ1 Where in the image does the evidence for a pedestrian come from, and which attribution maps can be trusted?

RQ2 How is "pedestrian" represented, and which finer pedestrian concepts are encoded beyond geometric confounds?

RQ3 What role does occlusion play, and how does the model detect pedestrians it cannot see?

RQ4 Does the planner use pedestrian queries, through which routes, and does its attention identify the ones that matter?

RQ5 Do the findings transfer to another dataset and camera rig (NAVSIM), and what does closed loop (CARLA) show?

Our contributions are:

1) An instance-level interpretability protocol for query-based E2E models: read-out points, query-to-annotation matching, safety strata, causal interventions that close every planner-side route (attention, collision rescoring, temporal memory) on cached perception, and scene-clustered statistics, implemented as a resumable pipeline built only from public code and data (Section IV).
2) A concept analysis of pedestrian perception combining class directions, control-task probes, TCAV, sparse autoencoders with specificity controls, and fine-grained concepts tested against their confounds with LEACE separability (Sections V-B and V-C).
3) A causal account of pedestrian reliance: attention points to the pedestrians that matter, but removing a pedestrian from attention and rescoring moves the plan by centimetres, and attention masking moves it less than for a matched car; group removal that also closes the memory route shows the same ordering, and a rare mode-switching tail carries the large effects (Section V-D).
4) Tests of the account on NAVSIM and in CARLA closed loop, including a Bench2Drive fine-tune with OpenDRIVE-derived map labels, and a test of detection at first appearance (Sections V-C and V-E).

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Fig. 1. One nuScenes validation frame (front-left, front, front-right cameras) with every annotated pedestrian coloured by the model's pedestrian probability p for the matched query; red = no query with maximum class score above 0.3 within 2 m; v = visibility level (1: 0–40 %, 4: 80–100 %). Pedestrians next to the parked truck are only weakly detected, one behind glass is missed, and on the right a fully visible pedestrian is missed between two detected ones.

II. Related Work

End-to-end driving and query-based perception. Planning-oriented models such as UniAD [12], VAD [13] and SparseDrive [11] share one network for detection, tracking, mapping, motion prediction and planning [14]. SparseDrive builds on the sparse, temporally propagated queries of the Sparse4D family [15], [16] and of StreamPETR [17], which sample image features with deformable attention [18]. Open-loop planning metrics on nuScenes can be matched by models that use ego status alone [19], [20], a shortcut that also bounds how much any perceived agent can matter to the plan; imitation learning is prone to such causal confusion [21], [22], and end-to-end CARLA agents carry comparable hidden biases [23].

Explaining driving models. Early work visualised attention of image-to-steering networks [24], and PlanT reads the attention of an object-level planner to identify the relevant agents [25]; recent models are trained for sparse, localised feature maps [26]. Object-level counterfactual explanations edit scene objects in image space [27], [28]; counterfactual audits remove or re-light objects and read the plan change [8], [9] or audit reliance on spurious cues [10]. Closest to us, Motzkus and Bernhard learn sparse dictionaries on the latent space of a trajectory-scoring planner and intervene on concepts to improve the NAVSIM score [7]. We work one level earlier, on the perception queries the planner reads, focus on pedestrians, control concepts for confounds, and close every planner-side route when testing reliance.

Concepts, units and interventions. Linear probes [29] with control tasks [30] test decodability; TCAV [31] tests whether a direction moves a class logit; LEACE [32] erases a concept linearly. Unit-level concept discovery goes back to Network Dissection [33] and ACE [34]; sparse autoencoders (SAEs) decompose superposed features [35] into sparse units [36], [37], archetypal variants stabilise them [38], and their units can depend on the random seed [39]. Our planner interventions are activation patching in the sense of causal mediation analysis [40], [41]; patching and steering along a subspace can produce effects through off-manifold directions [42], which is why we pair steering with norm-matched random directions. Whether attention explains a decision is contested [43], [44]; we test it causally for a planner. Probing has shown that 2D detection transformers encode 3D object properties [45].

Evaluating attribution. Gradient saliency [46], SmoothGrad [47] and D-RISE [48] are evaluated with ROAD [49] and model-randomisation sanity checks [50].

TABLE I
Positioning against the closest recent work. ✓ yes, (✓) partly, – no. CVAA analyses representations after input edits but not concepts, and selects objects but not pedestrians specifically.

	inside perception	pedestrian focus	concept discovery	confound control	closed loop
Motzkus & Bernhard [7]	– (planner)	–	✓	–	(✓)
Yang et al. [8]	– (input edits)	✓	–	–	–
CVAA [9]	(✓)	(✓)	–	–	–
Ours	✓	✓	✓	✓	(✓)

III. Model and Read-Out Points

We analyse the released SparseDrive-S stage-2 checkpoint [11] without modification. On nuScenes validation [51] it gives NDS 0.526, mAP 0.414, map mAP 0.566, AMOTA 0.370, planning L2 0.61 m and collision rate 0.10 %, within 0.01 of the values reported in the authors' repository. Figure 2 shows the parts we read and intervene on. A ResNet-50 with FPN encodes six cameras; the detection decoder refines 900 object queries over six layers, each sampling image features around the query's 3D anchor with deformable attention and attending to the other queries and to the instance bank; 600 queries are carried to the next frame (the detector's memory). The planner prepends an ego query to the 50 highest-scoring agent queries; in each planner layer the agents and the ego query attend to each other, to map queries and to a temporal queue holding their own features of the previous three frames (for the ego query, its post-planner features). The planner outputs six trajectory modes per driving command; a collision-aware rescoring penalises modes whose path overlaps the predicted motion of any detection with score ≥ 0.5, and the best remaining mode is returned (3 s, 6 waypoints).

IV. Method

A. Instances, matching and strata

Each nuScenes pedestrian annotation within 50 m with at least one lidar or radar point is matched to at most one query by Hungarian assignment on the BEV centre distance (threshold 2 m) among queries whose maximum class score exceeds 0.3. Matching is class-agnostic, so a matched pedestrian may be classified as another class. The validation split contains

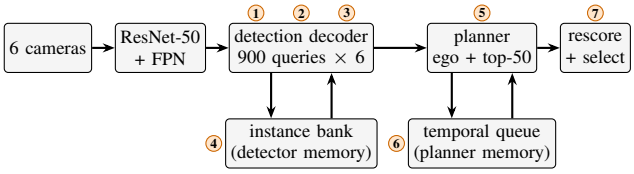


Fig. 2. SparseDrive and our read-out points: **1** deformable sampling (attribution); **2** per-layer query features (probes, TCAV, SAEs, concepts); **3** class head (class directions); **4** detector memory (history ablation); **5** planner attention (masking, dropping, feature replacement, relocation); **6** planner memory; **7** collision rescoring. Routes 5–7 are closed together in the complete-removal experiment.

26,770 pedestrian annotations, of which 0.59 are matched. Strata: *visibility* (nuScenes levels 1–2 = 0–60 %, 3–4 = 60–100 %), *distance* (<15, 15–30, >30 m) and *corridor*: ahead of the ego vehicle, within 8 m of the expert future path and with a constant-velocity time-to-collision below 8 s (1,953 pedestrians). The corridor uses the expert's future path and is therefore an offline stratification; a stricter variant (2.5 m, 4 s) contains 37 pedestrians.

B. Notation

For a plan $\hat{\tau} \in \mathbb{R}^{6\times 2}$ and the unmodified plan τ of the same frame, the plan change is $\Delta L_2 = \frac{1}{3}\sum_{t\in\{1,2,3\,\mathrm{s}\}} \|\hat{\tau}_t - \tau_t\|_2$; we also report the longitudinal change at 3 s, whether the selected mode switches, and the change of the minimum clearance to the annotated future pedestrian boxes. For a concept direction v and a pedestrian with layer-l feature h and final pedestrian logit z, TCAV counts the fraction of pedestrians with $\nabla_h z \cdot v > 0$. Rank-AUC treats the agents of a stratum whose masking effect is in the top 20 % as positives and pools agents across frames. The steering strength α scales a unit pedestrian direction to the mean norm of the top-10 pedestrian units' contribution (Section V-B). Several "pedestrian effects" are reported and differ by population and intervention: all detected pedestrian agents under attention masking (mean over every agent); GT-matched pedestrians in the matched comparison with cars; corridor pedestrians under attention masking, rescoring removal or both; and whole-group removal per frame.

C. Attribution (*RQ1*)

For a matched query we compute five maps of its pedestrian logit over the six images: the deformable-attention sampling density (DFA), the input gradient [46], SmoothGrad [47] (n=16), gradient×feature on the FPN features, and D-RISE [48] with 500 masks and a mean-centred estimator (plain RISE sums are dominated by a constant and by mask-density noise). Faithfulness uses ROAD [49], with the noisy linear imputation approximated by Jacobi diffusion to the same harmonic fixed point plus Gaussian noise (GPU-batched). Randomisation cascades from the last refinement layer towards the backbone [50]; a method fails if its map at the last stage with finite activations keeps a Spearman or SSIM similarity above 0.7. Label randomisation is replaced by a target-randomisation proxy (the map for another class), run for the gradient and gradient×feature.

D. Concept analysis (*RQ2*, *RQ3*)

Class directions and probes. We compare the final classifier's pedestrian weight with the other classes and record which class wins for each matched pedestrian. Logistic-regression probes on layer-2, 4 and 6 query features are trained on training scenes and tested on validation scenes (balanced accuracy). Selectivity is the plain accuracy above a control task in which every tracked object keeps one random label [30], computed on a random frame split because objects must recur.

TCAV. For a concept we fit 30 concept activation vectors [31] on bootstrap samples of training-split features and 30 random vectors on random partitions of the same pool, and take exact gradients of each validation pedestrian's final logit with respect to its own layer-l feature through all later decoder layers. Scores are compared with Welch's t-test, Bonferroni-corrected over 12 tests (four concepts, three layers). For occlusion we add two confound-controlled versions: occluded and visible sets matched cell by cell on distance and box height, and features with distance and box height (linear and squared) regressed out.

Sparse autoencoders. We train TopK SAEs [37] (2048 or 8192 units, $k \in \{16, 32, 64\}$) on 1,449,892 layer-6 query features of the training split (score > 0.1) and the dictionaries of the Overcomplete toolbox [38]. A dictionary is accepted if it explains >90 % of the variance with at most 64 active units and changes pedestrian AP by at most one point when its reconstruction is spliced into the live model (centre-distance AP averaged over 0.5/1/2/4 m on every fourth validation frame). Units are catalogued by the classes of their 32 strongest activations; a unit is pedestrian-specific if at least 24 of them are pedestrians. For the primary SAE (2048 units, k=64) we ablate the ten strongest pedestrian units, ten car units or ten random units, and steer non-pedestrian queries along the pedestrian direction or a norm-matched random direction; stability is tested by retraining with two further seeds.

Fine-grained concepts. From annotations, tracks and the nuScenes map we label ten contrasts among matched pedestrian instance-frames of both splits (Table VI). Each is tested with a probe on the features, a probe on its confounds alone (the concept's distance, speed, visibility, box height or lateral offset, with squares), a probe after regressing the confounds out of the features, and a shuffled-label control. A concept is encoded beyond its confounds if the lower 95 % scene-bootstrap bound of the residual probe exceeds the shuffled control (or 0.5) by at least five points; "weak" if it exceeds it by less. Separability uses LEACE [32]: erase concept A, re-probe concept B, and report B's retained accuracy above chance (self-erasure is not reported, since the re-fitted probe on erased features is ill-posed).

E. Planner interventions (*RQ4*)

The deformable-aggregation kernel is non-deterministic (atomic additions), so re-running perception changes plans. All planner interventions therefore re-run only the planner on cached perception, which makes the unmodified pass bit-identical. Single-agent interventions on agent query j: masking j as a key in the current frame's planner attention; removing

j from the rescoring; both; dropping j from the top-50; replacing its feature by the mean pedestrian feature or another pedestrian's feature; and moving its anchor 3 m into or out of the path (an off-manifold probe, since the query feature still encodes the original position). *Complete removal* closes all planner-side routes for a group (all detected pedestrians, vehicles, or agents with score > 0.3; queries below this threshold are not removed, so the test covers confidently detected agents): the group is masked from attention and removed from rescoring in every frame of the validation stream, and the masked pass is committed to a separate planner memory, so that the ego history never attended to the group. Perception-side mixing (detector self-attention, the image-derived ego feature) remains and is outside a planner-reliance claim. To test whether a cheap signal predicts reliance we rank agents by ego-query attention, by the plan gradient (a first-order estimate of the masking effect, used as an upper reference), by inverse time-to-collision (predicted agent velocity, constant ego velocity) and randomly.

F. Temporal ablation (*RQ3*)

The detector is re-run with its instance bank truncated to 0–3 frames of history ("full" = the unbounded recurrent bank), and with the detection head's temporal attention zeroed while keeping propagated queries. The first-appearance test bins validation pedestrians by the number of earlier keyframes in which the same instance was annotated; the first keyframe of each scene is excluded, since every pedestrian in it would count as newly appearing.

G. External validity (*RQ5*)

NAVSIM. We run the nuScenes checkpoint zero-shot on NAVSIM navtest logs [1] (camera mapping to the nuScenes rig, 3 s plans extended to 4 s at constant velocity) and score with the official one-stage EPDMS. Sensor data were available for 38 logs (selected by sensor availability on our mount, not by scenario content); inference ran on 1,934 of the 2,129 cached scenes, and after token filtering 1712 scenes from 33 logs are scored. Interventions: "masked"/"dropped from top-50" hide detected pedestrians in the last of the four history frames only (planner attention or top-50); the complete-removal variants act in every frame, on memory and on rescoring, for pedestrians, vehicles or all agents. A constant-velocity agent gives an ego-status reference.

CARLA closed loop. We fine-tune the checkpoint on 100 Bench2Drive clips [2], without map labels (map head frozen) and with map labels derived from the OpenDRIVE maps in SparseDrive's three classes (lane dividers de-duplicated, road boundaries as the contour of the drivable-area union, pedestrian crossings including those stored only as OpenDRIVE objects). All three models drive the 20 Bench2Drive routes with a pedestrian scenario (pedestrian crossing, object crossing from behind an obstacle, parked-car crossing, pedestrian during a turn), twice each, in CARLA 0.9.15 [52]. Statistics use the route as the unit (paired, 10,000 bootstrap resamples).

TABLE II
EVALUATED SUBSETS (NUSCENES V1.0-TRAINVAL; VALIDATION = 6,019 FRAMES IN 150 SCENES).

Analysis	Subset
Matching, strata, class directions	all frames, both splits
Probes	250,000 matched queries, both splits
SAE training / catalogue	1.45 M training queries / 0.5 M
Attribution	1 frame per validation scene (126 frames, 306 targets)
SAE splice, steering, controls	every 4th validation frame (1,505)
TCAV	every 3rd validation frame (4916 pedestrians)
Planner single-agent	every 3rd validation frame (2,007; 100,350 agents)
Complete removal	every validation frame (6,019); table on frames with the group
Temporal ablation	every 6th validation frame

H. Statistics and evaluated subsets

CIs are 95 % cluster-bootstrap intervals over scenes (1,000 resamples) unless stated: the matched pedestrian–car comparison uses 500 resamples, NAVSIM 2,000 over logs and CARLA 10,000 over routes. Table II lists the evaluated subset of every analysis; all use the full nuScenes v1.0-trainval split. Numbers are generated from the result files by a script, except the attribution table and a few counts noted under Reproducibility.

V. RESULTS

A. *RQ1*: Where the evidence comes from

Table III and Fig. 3 summarise 306 pedestrian targets. On corridor pedestrians, the DFA map, D-RISE and gradient×feature reach the lowest MoRF areas with overlapping CIs, so no single method is most faithful. LeRF is saturated near 1 for all maps except D-RISE, which falls below random on it. The methods disagree: pixel rank correlation is 0.18 between DFA and the gradient and 0.06 between the gradient and D-RISE, while the gradient family agrees internally (0.69). DFA fails randomisation: its sampling locations follow the anchor geometry, so it shows where the query looks rather than why it decides. SmoothGrad passes narrowly (last-stage SSIM 0.695 against a 0.7 threshold, within noise of Gradient's 0.740 fail); D-RISE's pass is degenerate, because after randomising the last refinement layer the reference detection no longer exists. Gradient maps are nearly identical for another target class, so they explain "an object is here" more than "a pedestrian is here". Qualitatively, the feature-based maps concentrate on the feet and ground contact point, where a camera-only model can read depth.

B. *RQ2*: How "pedestrian" is represented

A clean class direction. The pedestrian weight vector is nearly orthogonal to every other class (largest cosine 0.13, with construction vehicle). Among the 15,671 matched validation pedestrian instance-frames, 1.56 % are out-scored by another class, most often by barrier (training split: 0.69 %); since only 0.59 of validation pedestrians are matched, this statement

TABLE III
ATTRIBUTION ON CORRIDOR PEDESTRIANS (n=69; ROAD MORF ↓, LERF ↑, 95 % CI), LOCALISATION ON ALL 306 TARGETS (POINTING GAME, ENERGY PG) AND CASCADING RANDOMISATION (LAST VALID STAGE: +DET-SHARED; D-RISE: BLOCKS 1–6).

Method	MoRF	LeRF	PG	EPG	MPRT
DFA	0.47 [0.40, 0.53]	0.98	0.82	0.54	fail
Gradient	0.64 [0.60, 0.69]	1.00	0.56	0.10	fail
SmoothGrad	0.60 [0.55, 0.64]	1.00	0.72	0.11	pass[a]
Grad×feat.	0.50 [0.43, 0.55]	0.99	0.66	0.15	fail
D-RISE	0.48 [0.44, 0.53]	0.88	0.49	0.03	degen.[b]
Random	0.96 [0.95, 0.97]	0.96	0.01	0.00	–

[a] last-stage SSIM 0.695 vs. threshold 0.7. [b] reference detection absent after randomisation.

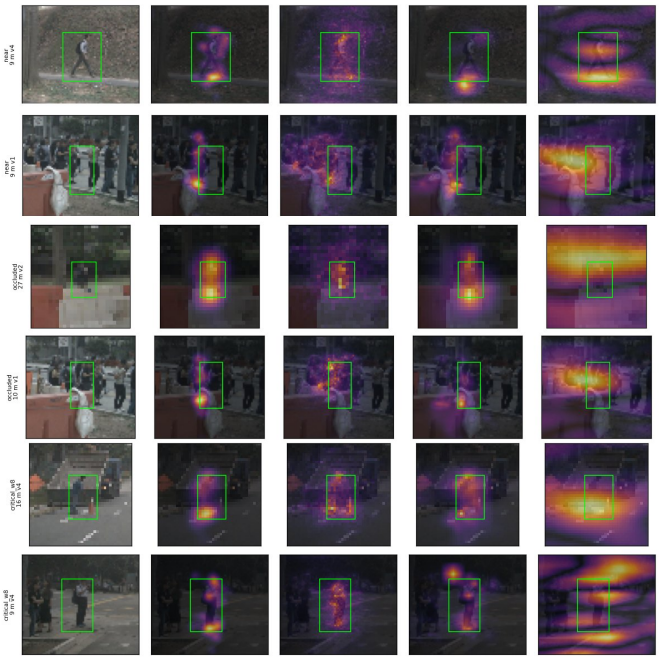


Fig. 3. Attribution maps for six validation pedestrians (rows: two near, two occluded, two in the corridor). Columns: input with annotated box, DFA, SmoothGrad, gradient×feature, D-RISE (the plain gradient is close to SmoothGrad and omitted). Banded D-RISE patterns are typical of a finite mask budget.

covers the pedestrians the model localises. Probes read the class from layer 2 on (Table IV). Distance and motion are decoded with similar balanced accuracy, whereas the four-level visibility is weakly decodable. Corridor membership is highly imbalanced (majority share above 0.98 in the strict corridor), so its plain-accuracy selectivity is capped and not informative.

Pedestrian units. The primary SAE explains 0.978 of the variance with 64 active units (2.1 % dead). 10 dictionaries meet all three acceptance criteria (Table V), including plain PCA, so reconstruction quality alone does not establish concepts. 20 of 2,002 live units are pedestrian-specific; in the 8192-unit SAE, 57 of 7,669 are (0.74 % vs. 1.00 %). Individual units are not stable across retraining: with two further seeds the pedestrian units are matched with mean cosine 0.61 (all units: 0.58), and

TABLE IV
LINEAR PROBES ON QUERY FEATURES, VALIDATION SCENES (BALANCED ACCURACY, %; CHANCE = 100/K). SELECTIVITY: PLAIN ACCURACY ABOVE THE CONTROL TASK, FRAME SPLIT. PROBE SET: 250,000 QUERIES.

Property (k classes)	layer 2	layer 6	chance	selectivity
object class (10)	90.2	90.2	10.0	55.4
distance bin (3)	91.6	89.8	33.3	47.4
visibility level (4)	42.1	41.4	25.0	8.4
in widened corridor (2)	84.1	83.6	50.0	7.0
in strict corridor (2)	71.8	79.1	50.0	0.4
moving (2)	91.9	92.5	50.0	22.5
TTC $<$ 4 s (2)	82.6	83.9	50.0	5.6

only 1 of 20 has a counterpart above 0.9 in both seeds (0 of the three pedestrian units shown in Fig. 4), in line with [39]. The concept is more stable than its units: every seed has a similar number of pedestrian units (20/20/19), their spans overlap with mean principal-angle cosine 0.62 (random unit sets of the same size: 0.31), and the pedestrian steering direction agrees with cosine 0.63. The units in Fig. 4 are therefore examples from one seed, not identified features; in this seed they are organised by viewing condition rather than pose: dim or distant pedestrians, close full-body walkers, pedestrians standing in crowds.

Ablating the ten strongest pedestrian units costs -4.9 pedestrian AP points and -2.0 car points, and moves the plan by 22.4 cm; ablating ten car units costs -14.5 car and -2.1 pedestrian points (plan 21.3 cm), and ten random units cost 0.00 and -0.01 points (plan 0.0, 1.0, 2.2 cm for the three random sets). The splice changes the perception of all agents, so its plan effect is not a test of planner reliance on pedestrians. Each set hurts its own class most, and random units do nothing, so the effect is class-selective rather than a generic cost of removing ten units; the cross-class losses show that the strongest units also carry evidence shared between object classes. Car units are chosen by car selectivity, computed exactly like pedestrian selectivity. Steering non-pedestrian queries along the pedestrian direction raises their pedestrian logit by 1.41 at α=4 (a norm-matched random direction: -0.40), but the share classified as pedestrian rises only from 0.47 % at α=0 to 0.50 %: the units push the class decision in the right direction without determining it.

Fine-grained concepts. Table VI separates what the queries encode about a pedestrian from what its geometry implies. "Moving" is decodable but explained both by the annotated speed and by the model's own predicted speed (residual 0.53 [0.50, 0.55]): motion is represented through the velocity estimate. 8 of the ten concepts are classed as beyond their confounds, among them walking towards the ego path, standing on a zebra crossing, being on the road, being in a group, construction worker and emerging from occlusion; sitting is weak, and child and night rest on few validation people (18 and 7) and are preliminary. The confound sets are geometric; for walking towards the path the predicted velocity direction, and for road and zebra map-derived distances, would be stricter controls that we have not run. Erasing any concept with LEACE keeps at least 75 % of every other concept's above-chance accuracy, so they are separable. Being in a group

TABLE V
DICTIONARIES ON LAYER-6 QUERY FEATURES: EXPLAINED VARIANCE, ACTIVE UNITS, CHANGE OF PEDESTRIAN AP WHEN SPLICED INTO THE LIVE MODEL, ACCEPTANCE.

Dictionary	EV	L0	ΔAP_{ped}	pass
sae_n2048_k64	0.978	64	-0.01	✓
sae_n8192_k64	0.978	64	-0.01	✓
sae_n2048_k32	0.927	32	-0.02	✓
OC-TopK_n2048_k64	0.975	64	-0.04	✓
OC-TopK_n8192_k64	0.975	64	-0.07	✓
OC-BatchTopK_n2048_k64	0.976	63	-0.05	✓
OC-MP_n2048_k32	0.968	32	-0.04	✓
OC-OMP_n2048_k32	0.971	32	-0.06	✓
OC-RATopK_n2048_k64	0.933	64	-0.26	✓
OC-PCA_64	0.947	64	-0.04	✓
OC-DictLearning_512	0.985	122	-0.01	✗
OC-ArchetypalAnalysis_256	0.389	4	-7.31	✗
OC-KMeans_512	0.585	1	-2.36	✗

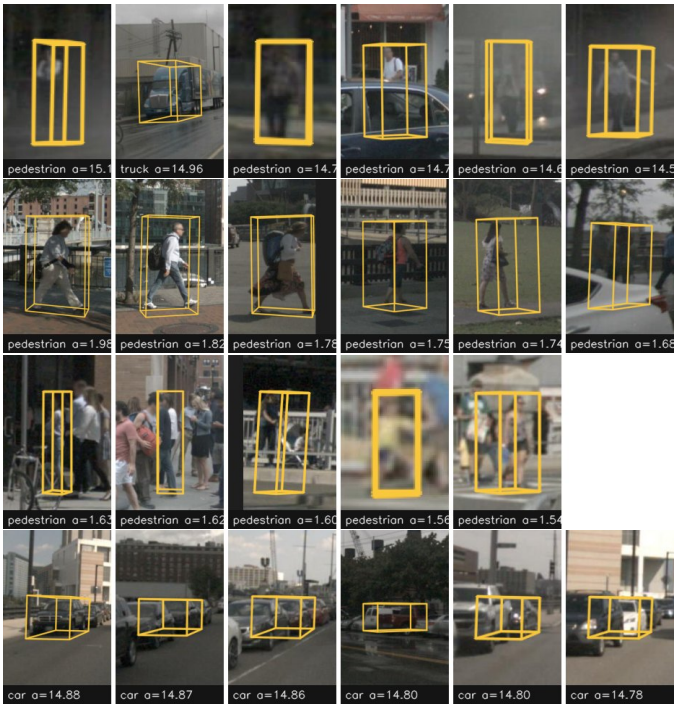


Fig. 4. Strongest activations of three pedestrian SAE units (rows 1–3: dim or distant pedestrians, with one truck among the top activations; close full-body walkers; pedestrians standing in a crowd) and one car unit (row 4). Training-set objects; a = activation.

pushes the pedestrian logit up and moving and sitting push it down; walking towards the path is encoded but does not move the class logit.

C. *RQ3: Occlusion and memory*

Occlusion is negative evidence. In the last decoder layer the occluded-vs.-visible direction lowers the pedestrian logit: TCAV 0.18 vs. 0.52 for random directions (Bonferroni $p = 8 \times 10^{-8}$, Table VII). The effect survives confound control: 0.19 vs. 0.49 with sets matched on distance and box height, and 0.16 vs. 0.54 with both regressed out. The direction reverses across layers (0.60 at layer 2), as does the corridor direction, which pushes towards the class in the last layer (0.74 vs. 0.50),

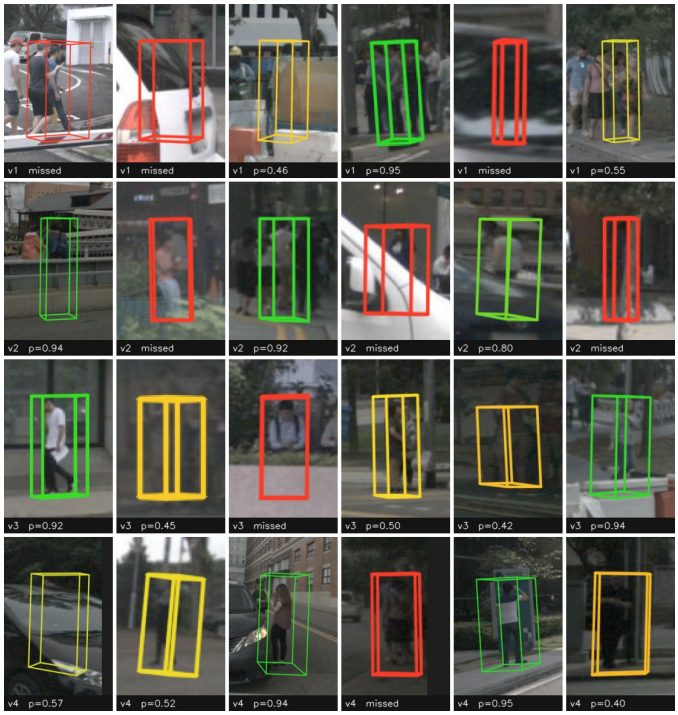


Fig. 5. Validation pedestrians closer than 30 m, one row per visibility level (top: 0–40 %, bottom: 80–100 %), coloured by pedestrian probability; "missed" = no query above 0.3 within 2 m.

a bias in the safe direction. Together with the weak visibility probe this suggests that occlusion acts as evidence against the class rather than being stored as an explicit value (Fig. 5).

Hidden pedestrians are found through memory. Without history, recall falls from 0.47 [0.41, 0.52] to 0.20 for partly or heavily occluded pedestrians (visibility 0–60 %, n = 1,586) and from 0.66 to 0.46 for visible ones: a relative loss of 58 % vs. 29 %, with a difference in absolute loss of 0.08 [0.03, 0.12]. Recall rises monotonically with 0–3 frames of history (Fig. 6). Zeroing the detection head's temporal attention barely changes recall (0.45), so the memory is carried by the propagated queries. Newly appearing pedestrians are rarely detected: excluding the first keyframe of each scene, where a pedestrian is "first seen" only because the scene starts, recall at first sighting is 0.11 for heavily occluded and 0.22 for visible pedestrians, and it rises to 0.55 and 0.71 after four or more earlier sightings. The penalty of a first sighting is not specific to occlusion: the occlusion × first-sighting interaction is +0.05 [-0.05, +0.16] (scene bootstrap; -0.08 [-0.18, +0.04] including scene starts).

D. *RQ4: Does the planner use pedestrians?*

Attention ranks the agents whose masking moves the plan. For corridor pedestrians (8 m / 8 s stratum), ego-query attention ranks the agents whose masking matters most with rank-AUC 0.90 [0.86, 0.93]; in the pre-specified strict stratum (2.5 m / 4 s) only 12 agents qualify and the AUC is uninformative (0.59 [0.17, 0.92]). Target and ranker share the attention pathway, so this AUC measures consistency between attention and its own removal rather than independent relevance; the

TABLE VI

FINE-GRAINED PEDESTRIAN CONCEPTS. n_{POS}: MATCHED INSTANCE-FRAMES OF BOTH SPLITS. BALANCED ACCURACY ON VALIDATION SCENES OF PROBES ON THE FEATURES (FEAT), THE CONFOUNDS ALONE (CONF) AND THE FEATURES WITH CONFOUNDS REGRESSED OUT (RESID). TCAV ON THE PEDESTRIAN LOGIT, * SIGNIFICANT AFTER BONFERRONI CORRECTION; THE RANDOM-DIRECTION MEANS LIE BETWEEN 0.46 AND 0.57. VERDICT: STRICT CRITERION (SECTION IV).

Concept (positive vs. negative)	n_{pos}	feat	conf	resid	TCAV	verdict
moving vs standing	50,452	0.84	0.94	0.50	0.29*	not beyond
sitting/lying vs standing	4,644	0.85	0.95	0.62	0.26*	weak
construction worker vs adult	3,205	0.81	0.73	0.76	0.52	beyond confounds
crossing towards path vs alongside	4,152	0.85	0.59	0.82	0.36	beyond confounds
on road vs walkway	9,442	0.80	0.56	0.80	0.37	beyond confounds
on zebra crossing vs walkway	5,324	0.85	0.58	0.83	0.58	beyond confounds
in a group vs alone	33,432	0.84	0.67	0.80	0.84*	beyond confounds
emerging from occlusion vs visible	4,405	0.68	0.54	0.67	0.60	beyond confounds
child vs adult	673	0.68	0.81	0.63	0.44	beyond confounds
night vs day	5,375	0.94	0.56	0.93	0.54	beyond confounds

TABLE VII

TCAV ON THE PEDESTRIAN LOGIT (n = 4916 VALIDATION PEDESTRIANS); BONFERRONI OVER ALL 12 TESTS; THE STRICT CORRIDOR HAS FEW POSITIVES.

Concept	layer	TCAV	random	$p_{Bonf.}$	sig.
corridor strict	2	0.42 ± 0.08	0.53 ± 0.14	1.3e-2	**yes**
corridor strict	4	0.29 ± 0.10	0.50 ± 0.18	3.5e-5	**yes**
corridor strict	6	0.61 ± 0.07	0.50 ± 0.18	5.9e-2	no
corridor w8	2	0.25 ± 0.04	0.48 ± 0.15	6.6e-8	**yes**
corridor w8	4	0.43 ± 0.06	0.49 ± 0.14	3.5e-1	no
corridor w8	6	0.74 ± 0.03	0.50 ± 0.22	2.4e-5	**yes**
near	2	0.37 ± 0.03	0.57 ± 0.16	1.2e-6	**yes**
near	4	0.15 ± 0.01	0.49 ± 0.19	1.2e-9	**yes**
near	6	0.58 ± 0.01	0.52 ± 0.18	9.9e-1	no
occluded	2	0.60 ± 0.08	0.50 ± 0.14	2.2e-2	**yes**
occluded	4	0.30 ± 0.05	0.41 ± 0.18	2.3e-2	**yes**
occluded	6	0.18 ± 0.04	0.52 ± 0.23	7.9e-8	**yes**

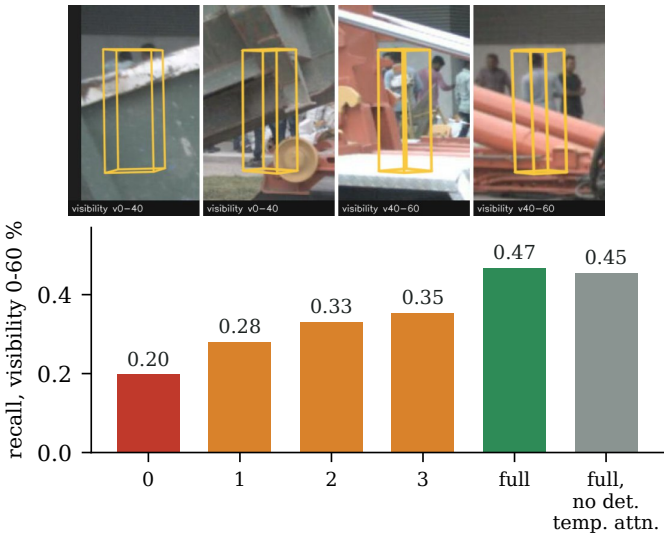


Fig. 6. Top: a pedestrian behind construction machinery over four keyframes (0.5 s apart); in the last frame it is detected only with ≥3 frames of history. Bottom: recall of heavily occluded pedestrians vs. history length.

plan gradient, a first-order estimate of the target, reaches 0.96. Inverse time-to-collision ranks them worse than chance in this stratum (AUC 0.34 [0.28, 0.41]): more urgent corridor pedestrians move the plan less (Spearman -0.34 between 1/TTC and the masking effect), so geometric urgency does not explain which pedestrians the planner uses. Randomising the planner lowers attention's AUC from 0.87 to 0.56 over all agents and from 0.87 to 0.63 over pedestrians.

Single pedestrians: attention and rescoring. For the corridor pedestrians, masking from current attention moves the plan by 3.6 cm [2.5, 4.9], removing from rescoring by 0.5 cm, and both by 3.8 cm [2.6, 5.2] (n = 1,549). The rescoring route is rarely active: in the existing rescore ablation 18 of 4,554 pedestrian agents change any penalty and 5 change the selected mode. Over all pedestrians the effect is 1.3 cm and flips the collision outcome in 1 of 7,555 cases. Relocating a pedestrian 3 m into the path (1.4 cm) and swapping its feature (1.3 cm) move the plan by similar small amounts.

For scale, the planner's own average L2 error to the expert trajectory is 0.61 m, so the mean single-pedestrian effect (0.72 cm) is about 1 % of that error. We do not test the normative question of whether the planner should react more, e.g. on frames where the expert yields to a pedestrian.

Encoded concepts are not used preferentially. Table VIII links each validation pedestrian to its planner row by position and compares, within distance bins, the attention percentile and the masking effect of pedestrians with and without a concept. Walking towards the ego path, being on a zebra crossing and being on the road, all encoded beyond their confounds (Table VI), receive neither more attention nor a larger masking effect; emerging from occlusion and construction workers receive less attention, and pedestrians in groups move the plan less.

Pedestrians vs. vehicles. By predicted class, the mean single-agent effect is 0.72 cm for pedestrians against 2.05 cm for cars and 4.08 cm for buses, and the pedestrian median (0.11 cm) is the lowest of all classes. Matching GT pedestrians and GT cars on distance, lateral offset to the path, detection confidence and corridor membership (69 strata, 6,123 pedestrians, 14,449 cars), a pedestrian moves the plan by 1.51 cm and a car by 3.72 cm, a difference of -2.21 cm [-2.78, -1.60]. Group removal shows the same ordering (Table IX): on the 4,283 frames with at least one detected pedestrian, removing all of them from attention, rescoring and memory

TABLE VIII
CONCEPT → PLANNER ON VALIDATION PEDESTRIANS: POSITIVE MINUS NEGATIVE ATTENTION PERCENTILE AND MASKING EFFECT (CM), AVERAGED OVER DISTANCE BINS (0–15, 15–30, 30–100 M); SCENE-BOOTSTRAP 95 % CIS. n_+ = POSITIVE PEDESTRIAN-FRAMES.

Concept	n_+	Δ attention pct.	Δ effect (cm)
moving vs standing	3452	-0.03 [-0.06, +0.01]	+0.13 [-0.57, +0.81]
sitting vs standing	265	-0.06 [-0.13, +0.01]	-0.99 [-1.77, -0.34]
worker vs adult	169	-0.14 [-0.20, -0.05]	-1.01 [-1.70, -0.35]
crossing vs alongside	273	-0.01 [-0.08, +0.06]	+0.92 [-1.46, +5.39]
road vs walkway	740	+0.05 [-0.00, +0.09]	+0.22 [-0.72, +1.11]
crossing zone vs walkway	411	-0.00 [-0.06, +0.06]	+0.12 [-1.18, +2.08]
group vs alone	2438	-0.06 [-0.10, -0.03]	-1.12 [-2.14, -0.35]
emerging vs visible	328	-0.12 [-0.17, -0.08]	-0.21 [-1.20, +0.89]
child vs adult	31	-0.02 [-0.19, +0.14]	-1.03 [-1.66, -0.13]
night vs day	24	+0.31 [+0.08, +0.38]	-0.14 [-1.22, +0.51]

TABLE IX
GROUP REMOVAL OVER THE VALIDATION SPLIT: PLAN CHANGE ΔL_2 (CM, SCENE-BOOTSTRAP 95 % CI) ON THE FRAMES WHERE THE GROUP IS DETECTED, WHEN THE GROUP IS MASKED FROM CURRENT-FRAME ATTENTION (CUR), ADDITIONALLY REMOVED FROM RESCORING (+RS), OR REMOVED FROM ATTENTION, RESCORING AND PLANNER MEMORY IN EVERY FRAME (COMPLETE).

Group	frames	cur	+rs	complete
pedestrians	4,283	5.2 [4.0, 6.5]	5.3 [4.1, 6.7]	5.6 [4.4, 6.9]
vehicles	5,882	31.9 [29.6, 34.4]	34.3 [31.4, 37.6]	35.5 [33.0, 38.6]
all agents	5,975	44.4 [39.3, 49.4]	47.4 [41.6, 53.7]	47.1 [42.3, 52.9]

in every frame moves the plan by 5.6 cm [4.4, 6.9], against 35.5 cm for vehicles and 47.1 cm for all agents, with on average 5.6 pedestrians and 11.9 vehicles per frame. Per removed agent this is about 1 cm for pedestrians and 3 cm for vehicles (a coarse normalisation, since group effects need not add up over agents). Closing the memory route adds little: complete removal changes the plan only slightly more than masking pedestrians in the current frame (5.2 cm), and in frames without a current pedestrian, where only the ego history differs, the change is 0.05 cm. In frames with a corridor pedestrian the effect of complete removal rises to 13.6 cm (mode switch in 3.7 %), and it changes the GT-collision outcome in 0.1 % of frames, against 1.6 % for vehicles.

The tail. Effects are skewed: the median single-pedestrian effect is 0.16 cm and the 99th percentile 17.1 cm, but 33 maskings move the plan by more than 0.5 m, up to 4.39 m. All effects above 74.5 cm involve a switch of the selected mode, which happens in 35 of 7,555 maskings (0.46 %; mean effect 0.91 m). Of these, 29 are GT-matched pedestrians at a median 14.5 m, 10 in the corridor, with median attention rank 6 of 50. Figure 8 shows the frame whose top-attended pedestrian is fully visible; it is the largest effect without a mode switch, not the largest effect overall.

*E. **RQ5**: External validity*

NAVSIM. Zero-shot, SparseDrive-S reaches EPDMS 0.577 on 1712 scenes from 33 navtest logs; 24.6 % of the scenes with a detected pedestrian contain an at-fault collision. Masking all detected pedestrians in the last frame changes the score on pedestrian scenes by at most 0.004, and the changed collision

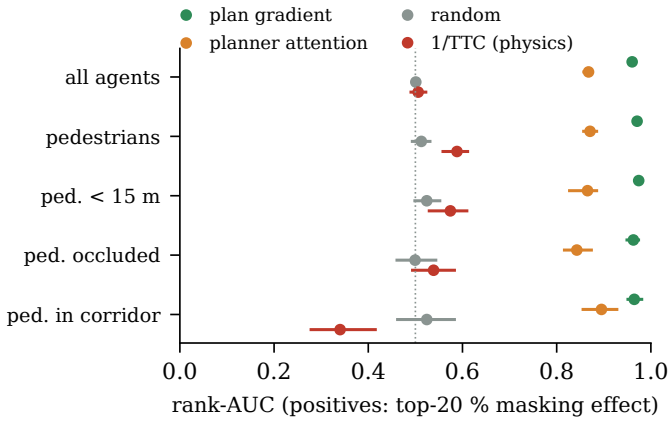


Fig. 7. Rank-AUC of four rankers for identifying the agents whose masking moves the plan most (scene-bootstrap 95 % CIs).



Fig. 8. A frame where the planner's top-attended agent of 50 is a fully visible pedestrian on a zebra crossing; masking it moves the plan by 74.5 cm without a mode switch, the largest such effect. Boxes are coloured by attention rank; "removal" = plan change when the pedestrian is masked.

outcomes (11 fixed, 4 new when masked; 13 and 11 when dropped from the top-50) are not significant (sign test $p = 0.12$ and 0.84). Table X adds complete removal (attention, rescoring and memory, in every history frame): EPDMS 0.578 with pedestrians removed, 0.554 with vehicles removed and 0.547 with all agents removed, against 0.577 unchanged and 0.312 for a constant-velocity agent. With paired per-scene differences and a log-clustered bootstrap (33 logs), removing vehicles lowers EPDMS by -0.023 [-0.036, -0.009] and the no-collision score by -0.027 (86 new and 40 avoided collisions), whereas removing pedestrians changes EPDMS by +0.001 [-0.006, +0.007] and the no-collision score by +0.005 [+0.000, +0.010] (5 new, 14 avoided). On a second dataset and camera rig, the planner's score therefore depends on vehicles; for pedestrians there is no evidence of degradation, and the no-collision shift towards fewer at-fault collisions is borderline (sign test $p = 0.064$; six group × metric comparisons, uncorrected). Without pedestrian detection recall on OpenScene, which we do not report, this null cannot separate weak reliance from missed detections.

CARLA closed loop. Table XI reports the 20 pedestrian routes. With the route as the unit, both fine-tunes complete less of the route than the zero-shot model (map labels: -32.0 points

TABLE X
NAVSIM ONE-STAGE EPDMS (ZERO-SHOT, 1712 SCENES). REMOVAL = GROUP REMOVED FROM PLANNER ATTENTION, RESCORING AND MEMORY IN EVERY HISTORY FRAME. PEDESTRIAN SCENES: $\geq$1 DETECTED PEDESTRIAN (1,382). NC = NO AT-FAULT COLLISION, EP = EGO PROGRESS.

	all	pedestrian scenes		
Planner input	EPDMS	EPDMS	NC	EP
unchanged	0.577	0.599	0.761	0.899
pedestrians removed	0.578	0.600	0.767	0.901
vehicles removed	0.554	0.579	0.732	0.900
all agents removed	0.547	0.571	0.725	0.902
constant velocity	0.312	0.333	0.611	0.818

TABLE XI
CARLA CLOSED LOOP ON THE 20 BENCH2DRIVE PEDESTRIAN ROUTES, TWO REPETITIONS EACH (40 RUNS PER MODEL). DS = DRIVING SCORE, RC = ROUTE COMPLETION (%), SUCC. = SUCCESS RATE (%); COLLISIONS ARE EVENT COUNTS OVER 40 RUNS.

Model	runs	DS	RC	succ.	ped.	veh.	layout
nuScenes checkpoint (zero-shot)	40	37.3	96.5	7.5	25	38	19
+ B2D fine-tune, no map labels	40	29.4	55.8	0.0	15	26	0
+ B2D fine-tune, map labels	40	37.8	64.7	12.5	15	29	2

[-45.3, -19.5]; no map labels: -40.9 [-52.9, -29.1]) and have fewer pedestrian collisions per run (-0.25 [-0.40, -0.12] and -0.25 [-0.42, -0.10]); driving-score differences are not resolved (map vs. zero-shot +0.8 [-13.3, +13.2]). A route stopped early meets fewer pedestrians, so the collision reduction is largely a consequence of driving less. The zero-shot model, which completes most routes, records 25 pedestrian collisions in 23 of its 40 runs; without a closed-loop removal run this does not show whether it reflects weak reliance or missed detections. Pedestrian interventions in closed loop are left for future work.

VI. DISCUSSION

Attention ranks, but does not measure, reliance. The planner's attention predicts which agents matter relative to each other, but not how much they matter: removing an attended pedestrian from attention and rescoring moves the plan by centimetres, and closing the memory route as well does not change this. Attention maps of a planner therefore cannot serve as evidence that a pedestrian was taken into account [43], [44]; a causal test that closes all planner-side routes can, and it is cheap on cached perception.

Pedestrians are used less than cars. At matched distance, path offset and confidence, and as a group, hiding pedestrians changes the plan less than hiding cars or vehicles. This is consistent with an imitation-trained planner on nuScenes, where the expert rarely yields to pedestrians, and with the ego-status shortcut [19], [20]. The large effects are mode switches, a discrete and rare mechanism that a safety argument has to characterise rather than average away.

The weak point is first appearance. The detector depends on its query memory, and a pedestrian that has just appeared has no memory yet: excluding scene starts, only 0.11 of occluded and 0.22 of visible pedestrians are detected at first sighting. Scenario-based testing (e.g. Bench2Drive's object-crossing routes, where the zero-shot model collides in most runs) should concentrate here.

Concept tests need confound control and data. "Moving" is decodable but explained by speed; without the confound-only and residual probes it would be reported as a learned concept. In a pilot on the two validation scenes of nuScenes-mini (not reported here), several concepts appeared confound-explained that the full data show to be encoded, so sample size is part of the protocol.

VII. LIMITATIONS

We analyse one model, SparseDrive-S; the protocol applies to other query-based planners and extending it is the most important next step. All nuScenes analyses are open loop, and the planner-reliance claims exclude perception-side mixing. Labels come from annotations, tracks and HD maps; concept tests are linear. The NAVSIM evaluation is zero-shot with a camera remapping on the 33 navtest logs with complete sensor data. The CARLA fine-tunes use 100 clips, closed-loop results vary between repetitions, and no pedestrian intervention was run in closed loop. Complete removal covers detections with score above 0.3; sub-threshold pedestrian evidence stays in the planner, and we do not report pedestrian detection recall on OpenScene. TCAV significance is computed over CAV refits against random CAVs, not over pedestrians or scenes, so it states that a direction is consistent, not how much the effect varies across the population.

VIII. CONCLUSION

Opening the object queries of an end-to-end driving model shows a pedestrian representation that is clean, spread over a subspace that is partly reproduced across seeds rather than by fixed units, sensitive to planning-relevant properties and supported by memory under occlusion, and a planner that attends to the right pedestrians but relies on them less than on cars. Interventions that close every planner-side route turn these observations into causal statements, which a safety argument for vulnerable road users needs.

REPRODUCIBILITY

The pipeline uses only the public SparseDrive code and checkpoint, nuScenes, NAVSIM/OpenScene, Bench2Drive and CARLA, and is resumable across interruptions (PyTorch 1.13.1, mmcv 1.7.1, flash-attention planner; eager and flash planner outputs agree to 10^{-3} m). Numbers and tables are generated from the result files by `build_numbers.py`; the attribution table (Table III) and a few in-text counts were transcribed from the generated report. Camera images are from nuScenes (Motional), used under CC BY-NC-SA 4.0. Code: https://github.com/pavanyadava007/seen-but-barely-used.

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